

SoilFER-LIMS Analytical Audit

Whether the GloSIS binding is real, and whether the analytical data model matches what a soil fertility laboratory actually does at the bench. Particle size analysis is the worked example; it is not the only one. Nothing was modified.

TARGET	DATE	REFERENCE
SoilFER-LIMS · lims.yigini.net soilfer-lims @ 6d830ac	1 September 2026	GloSIS v1.0 ontology network (glosis-ld/glosis); ISO 28258; local catalogue of 275 procedures

FINDINGS

34 across 7 domains · 8
affecting reported values

The GloSIS work is a catalogue, not a binding. The 275 procedures were fetched from the real ontology and are faithfully reproduced. But what the system stores against an analysis is a *procedure codelist group* — pSA, pH, exchangeableBases — labelled and treated as though it were the observed property. GloSIS separates those two things precisely, and the whole point of the ontology is that separation. Nothing the system exports carries a procedure URI, a QUDT unit, or a depth interval, so no GloSIS harvester could ingest it. Where an administrator has not chosen an attribute, the code *fabricates* a GloSIS URI from the analysis code and publishes it through the public SIS API — identifiers that look authoritative and resolve to nothing.

Particle size analysis exposes the deeper problem. One dispersion, one

sedimentation run, three fractions that must close to 100 % — modelled as three unrelated analyses, three assignable work items and three independent result rows, re-assembled in the browser for display. The closure check exists only in React, warns rather than blocks, and uses a ± 0.5 % tolerance that Bouyoucos hydrometry will fail routinely. No texture class is ever derived. No particle-size scheme (USDA 0.05 mm vs ISO 0.063 mm) is recorded, so silt and sand values are not comparable between the seven laboratories this platform is meant to harmonise.

The most serious findings are not about GloSIS at all. Every certificate states that results are reported on an oven-dry basis; nothing in the system measures moisture or applies the correction. The interpretation engine that prints fertiliser advice for farmers is keyed on parameter code alone — unit-blind and method-blind — and the shipped default unit (g/kg) disagrees with the thresholds it feeds (%). A soil at 2.0 % organic carbon is reported as *High — continue current practices* when it should read *Low — increase organic inputs*. That is advice going out over a signature.

8 CRITICAL Wrong numbers or wrong advice leave the laboratory	13 HIGH Data not defensible, traceable or harmonisable	13 MEDIUM Scientific completeness and fitness for purpose	6 SOUND Correct instincts to build the remediation on
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01 What GloSIS conformance actually requires

Worth stating precisely, because the gap between the implementation and the standard is a gap in *kind*, not in completeness.

GloSIS is a SOSA/SSN-based ontology aligned to ISO 28258. A soil observation in GloSIS is a four-part statement, and all four parts are mandatory for the statement to mean anything:

Feature of interest	GL_Layer / GL_Horizon, with upper and lower depth	0–20 cm layer of profile GTM-014	Absent — Sample has no depth field
Observed property	glosis_cl:ClayPropertyCode, SiltPropertyCode, SandPropertyCode — one per fraction	clay content	Conflated with the procedure group
Procedure used	sosa:usedProcedure → http://glosis.org/ont/procedure/pipette-disp	pipette method, dispersed	Stored but never emitted
Result with unit	glosis_lh:ClayValue, QUDT unit GM-PER-HectoGM	28 g/hg	Free-text string, free-text unit

The distinction that has been lost is the first-order one. In the ontology, Sand, Silt and clay are three separate observation classes, each with its own property code and its own Value result class, all in GM-PER-HectoGM (g per 100 g, i.e. per cent). GloSIS additionally defines ParticleSizeFractionsSum — an observation whose entire purpose is to carry the closure check, with the constraint that sand + silt + clay = 100 % — and TextureLabClass as a further, string-valued observation for the triangle class. pSA is none of these. It is the name of the *procedure codelist* that says *how* the distribution was determined: pipette, hydrometer, beaker or laser, crossed with dispersed / not dispersed / unknown, and again with a spectral variant. Twenty-five ways to do one thing — not a thing that can be measured.

The give-away is in the labels the system generates. The catalogue renders pSA as "**P S A**" and pH as "**P H**" — camel-case splitting applied to codelist class names that were never property names. The admin screen then tells the user these are "275 standard analytical procedures across **27 soil properties**". They are 275 procedures across 27 *procedure groups*. There is no clay in the list, because clay is not a procedure.

02

Critical — wrong values and wrong advice

These change what a laboratory reports to a farmer, a ministry or a research partner. Ordered by consequence.

ADV-01 CRITICAL

Fertiliser advice is generated unit-blind, and the shipped default unit is the wrong one

```
client/src/components/report/ReportContent.jsx:23-72 ·
client/src/components/admin/AnalysisManager.jsx:42
```

The interpretation engine is a lookup on parameter code and nothing else: `getInterpretation(item.param, item.value)` reads `THRESHOLDS[param]` and walks the bands. It never consults the unit the value was stored in, and never consults the method. The organic-carbon and nitrogen bands are expressed in per cent. The analysis configuration screen defaults new analyses to g/kg.

A laboratory that accepts the default and reports SOC in g/kg — which is the SI-correct choice, and what most soil laboratories now use — gets advice computed as though those numbers were percentages. The bands are 1 / 2 / 4, so any value above 4 lands in the top band:

PARAMETER	REAL SOIL	STORED	SYSTEM SAYS	SHOULD SAY
Organic carbon	2.0 %	20 g/kg	High — "Good organic matter content. Continue current practices."	Low / Medium boundary — "Increase organic inputs"
Total nitrogen	0.15 %	1.5 g/kg	High — "Sufficient nitrogen supply."	Low — "Consider nitrogen fertilization"
Exchangeable K	0.51 cmol(+)/kg	200 mg/kg	High — "Sufficient potassium supply."	Medium — adequate, monitor

In each case the advice is not merely imprecise, it is inverted: a deficient soil is certified as sufficient, and the farmer is told to apply nothing. This text is printed on the customer

report beneath a laboratory signature, and the same engine drives the "Recommendations" section, which filters on the interpreted level — so a genuinely deficient soil is also silently dropped from the recommendations list.

FIX

Interpretation must take the value, its unit and its method, normalise to the threshold set's own unit, and refuse to interpret when it cannot. Refusing is always acceptable; guessing is not.

BAS-01 CRITICAL

Every certificate claims an oven-dry basis the system never establishes

`client/src/components/report/ReportContent.jsx:346`

The report footer reads, on every certificate: *"Results are reported on an oven-dry weight basis unless otherwise indicated."*

There is no moisture determination anywhere in the system. No hygroscopic water field, no 105 °C residual moisture, no correction factor, no basis flag on the result. The technician weighs air-dry soil, enters a number, and the certificate asserts it is oven-dry. For a typical air-dry soil holding 2–5 % hygroscopic moisture — more in a clay, more in a humid climate — every mass-basis result on that certificate is biased low against its own stated basis by that amount, and no reader has any way to know. GloSIS carries `moistureContent` with nine procedures precisely because this correction is expected to be recorded.

This is the finding I would raise first in an accreditation assessment. It is an unsupported claim about the metrological basis of every number the laboratory issues, and it is trivially falsifiable.

FIX

Either measure and apply the correction — a moisture analysis, a stored factor, a basis field on each result — or remove the sentence and state the actual basis. Removing the sentence takes an hour and stops the system asserting something untrue; do that today and implement the correction properly in Phase B.

GLO-01 CRITICAL

The system mints fake GloSIS URIs and publishes them through the public API

`server/controllers/analysisController.js:159–160` ·
`server/controllers/sisController.js:122–126`

When an analysis is created without a GloSIS attribute chosen, the controller invents one

from the analysis code and constructs an ontology URI around it:

```
glosisAttribute: glosisAttribute || code,
glosisUri: glosisUri || `http://glosis.org/ont/glosis#${glosisAttribute || code}`,
```

The SIS export then applies the same fallback a second time, this time from the *result parameter name*:

```
uri: aMeta.glosisUri || `http://glosis.org/ont/glosis#${aMeta.glosisAttribute || r
```

So a locally-invented analysis code such as SAND, P_OLSEN or CE_1_5 becomes `http://glosis.org/ont/glosis#P_OLSEN` — a well-formed, authoritative-looking, entirely non-existent identifier, emitted under a "glosis" key to any consumer of the national Soil Information System API. Every laboratory invents its own codes, so every laboratory mints its own private namespace inside the FAO's. This is worse than having no binding: a downstream harvester cannot distinguish a fabricated URI from a real one, and the pollution propagates into whatever aggregate the SIS feeds.

FIX

Remove both fallbacks. An analysis with no GloSIS mapping must export `null` and be reported as unmapped. Then audit the `Analysis` table for URIs already fabricated and clear them.

GLO-02 CRITICAL

Procedure codelist groups are stored as if they were observed properties

```
server/config/glosis_catalog.json · server/prisma/schema.prisma:627 ·
client/src/components/admin/GlosisExplorer.jsx:63
```

The catalogue's 27 "attributes" are the GloSIS procedure codelist groupings, not properties. `Analysis.glosisAttribute` holds one of them, and both the UI and the SIS export present it as the property the analysis measures.

The consequence is a loss of resolution that cannot be recovered downstream. Sand, silt and clay are three distinct GloSIS properties; all three must be tagged `pSA`, because that is the only relevant entry in the catalogue. On export all three carry the same attribute and the same URI, and a consumer cannot tell which fraction a number represents. The same collapse hits every multi-analyte family: `exchangeableBases` covers Ca, Mg, K and Na; `extractableElements` covers 25 procedures across P, K and the micronutrients;

totalElements another 19. A single tag for each.

FIX

Two fields, not one: glosisProperty (from the property codelists in glosis_layer_horizon.ttl) and glosisProcedure (from the 275 already catalogued). The property belongs on the analysis; the procedure belongs on the methodology, where it already sits.

GLO-03 CRITICAL

No procedure is ever exported, so no observation is GloSIS-valid

server/controllers/sisController.js:115–130 ·
server/prisma/schema.prisma:642–658

The Methodology model carries glosisProcedure, glosisNotation, glosisDefinition, glosisReference, glosisCitation and glosisUri — the procedure URI that sosa:usedProcedure requires. None of it is read at export. The SIS payload emits attribute, methodLabel, definition, citation and the property URI — the procedure URI is not among them. The entire GloSIS procedure block is write-only.

Worse, the label that *is* emitted is wrong in a specific and predictable way. The export reads `r.method || aMeta.methodLabel`; Result has no method field, so the first term is always undefined and the second is the *analysis-level default*. A laboratory with two methodologies for pH — in water and in KCl — exports whichever was configured as the analysis default, regardless of which was actually run on that sample.

FIX

Record the methodology on the result (see TRC-01), then emit its procedure URI as usedProcedure. Until the result knows its own method, no export from this system can be trusted at the method level.

TRC-01 CRITICAL

A result cannot be re-analysed without destroying the original

server/prisma/schema.prisma:596–608 · resultsController.js:72 ·
workbenchController.js:491

The Result table is unique on (sampleId, param) and both write paths are prisma.result.upsert. One row per parameter per sample, forever. Writing a new value overwrites the old one in place, with no version, no supersession, no history.

The workflow explicitly supports re-analysis — WorkItem.reanalysisReason and

reanalysisRequestedBy exist, and managers can request it from the review queue. So the system invites a re-run and then erases the evidence of the first one. When a manager rejects a value as suspect and the technician re-runs it, the record afterwards shows only the second number, as though the first had never existed. There is no way to answer "what did we originally report, and when did it change?" — which is the first question in any client dispute or audit.

The same constraint makes analytical duplicates structurally impossible to store. A laboratory running duplicates at the standard 1-in-10 or 1-in-20 rate — as every accredited soil laboratory does — has nowhere to put the second determination.

FIX

Make `Result` append-only: add `replicateNo`, `supersededBy`, `isCurrent`, and drop the unique constraint to (`sampleId`, `param`, `methodologyId`, `replicateNo`). This is the single change that most improves the defensibility of the data.

ADV-02 CRITICAL

Phosphorus and salinity advice ignores the extraction and the extract ratio

`client/src/components/report/ReportContent.jsx:44–61`

There is one AVAIL_P band set — 5 / 15 / 25 mg kg⁻¹ — and it is applied to whatever the laboratory measured. But available P has no meaning independent of its extractant. The catalogue offers 25 extractable-element procedures precisely because Olsen (0.5 M NaHCO₃, pH 8.5), Bray-1 (dilute HCl/NH₄F, acid soils) and Mehlich-3 do not measure the same pool and do not share a calibration. A value of 25 mg kg⁻¹ is comfortably adequate by Olsen and at or below the critical range by Mehlich-3. The system prints "Adequate for most crops" either way. The programme spans Guatemala, Honduras, Zambia, Ghana, Kenya, Mozambique and Tunisia — acid tropical soils where Bray or Mehlich is appropriate, and calcareous soils where only Olsen is valid. One band set cannot serve both.

The EC bands have the same defect in a sharper form. The thresholds — 2 / 4 / 8 dS m⁻¹, "Non-Saline" to "Strongly Saline" — are the classical *saturated paste extract* (EC_e) scale. Many laboratories, including most running routine fertility work, measure EC in a 1:2.5 or 1:5 soil:water suspension, where readings are lower than EC_e by roughly a factor of 3 to 10 depending on texture and gypsum content. A 1:5 reading of 0.5 dS m⁻¹ can correspond to an EC_e of 3–5 dS m⁻¹ — moderately saline, yield loss in sensitive crops — and is certified "Non-Saline. No salinity concerns."

FIX

Band sets keyed on property *and* procedure, with an explicit extract-ratio field for EC.

Where no calibrated band set exists for the method in use, print the value and no interpretation.

PSA-01 CRITICAL

The sum-to-100 constraint exists only in the browser, and does not block anything

client/src/pages/TechWorkbench.jsx:21, 1062–1076, 1367–1408

The workbench computes `getTextureSum()` and marks the row valid when all three fractions are present and the sum is within $\pm 0.5\%$ of 100. The warning row reads *"Sand + Clay + Silt = 97.8 % — must equal 100 % ($\pm 0.5\%$) to complete"*.

It does not. `texSum.valid` is used for a badge colour and that warning line. It is not referenced in `buildEntries`, not in the save handler, not on the Complete button. The three values save and the work items complete regardless. The server knows nothing about texture at all — `validateResult` checks only a per-analysis min and max, one parameter at a time, so a set of fractions summing to 87 % or 113 % passes every server-side check and lands in `Result`. Any other write path — the API, a bulk import, a future spectral prediction — has no check whatsoever.

GloSIS models this constraint as a first-class observation, `ParticleSizeFractionsSum`, with the procedure `SumTxtr_calcul` ("Calculated sum of sand, silt and clay fractions") for exactly this purpose. The concept is available and unused.

FIX

Move closure to the server as a composite-analysis rule evaluated on save, and store the computed sum as its own result against `SumTxtr_calcul`. The browser check stays as fast feedback, not as the control.

03 Particle size analysis, end to end

The user named this example, and it is the right one: it is where the difference between a bench procedure and a database row is most visible.

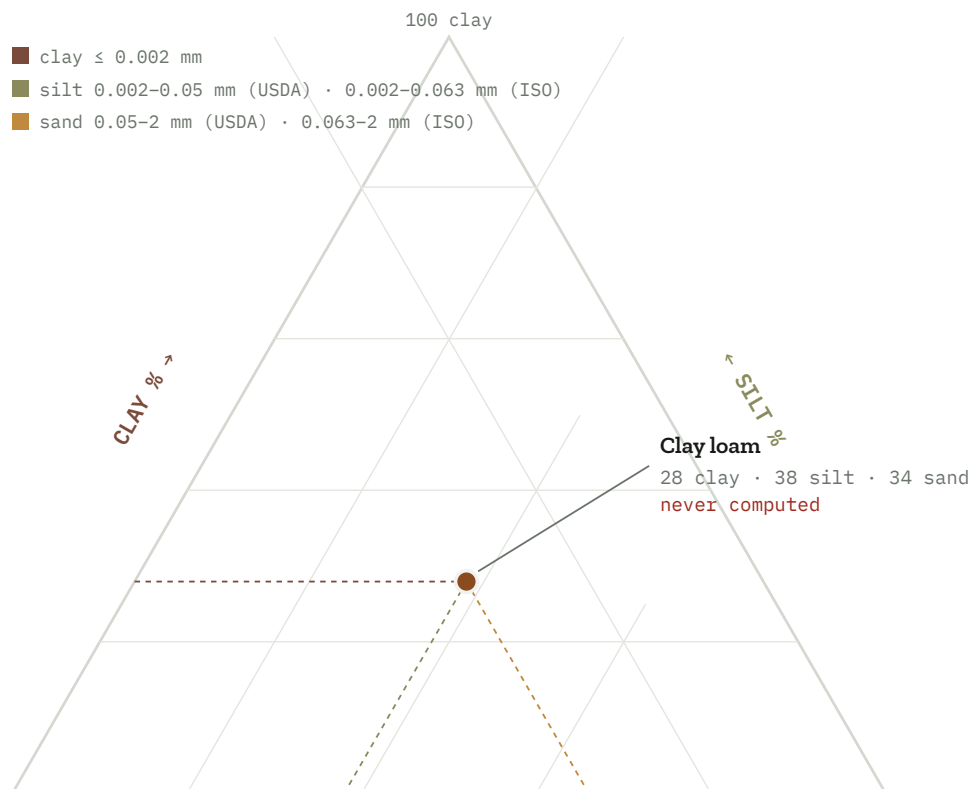
What happens at the bench

A pipette or hydrometer determination is *one procedure on one aliquot*. The technician weighs the fine earth, destroys organic matter and carbonates if the pretreatment calls for it, disperses with sodium hexametaphosphate, and then takes sedimentation readings against Stokes' law at the times that correspond to the fraction boundaries. Sand is recovered by wet sieving and weighed. Clay comes from the last withdrawal. Silt is very often obtained *by difference* — 100 minus sand minus clay — which is precisely why the three numbers are not independent measurements and why closure is a property of the method rather than a coincidence to be checked afterwards.

What the system does

Three separate Analysis records — SAND, SILT, CLAY — which generate three separate WorkItem rows, individually assignable, individually completable, individually equipment-logged, individually batchable for QC. The React workbench (TechWorkbench.jsx:310) then reassembles them into a virtual "TEXTURE" group for display, keyed by sample id.

The instinct behind that merge is right and worth keeping. But it is a presentation-layer repair of a model-layer problem, and everything below the presentation layer still sees three unrelated measurements: the three work items can be assigned to three different technicians on three different days; each can be completed alone; the manager review queue treats them as three independent decisions; and the audit trail records three procedures where one was performed.



Schematic USDA triangle. The system holds the three co-ordinates and never names the point. Note also the two silt/sand boundary conventions in the key — the system records neither, which is what makes cross-laboratory silt values incomparable.

PSA-02 HIGH

No particle-size scheme is recorded, so silt and sand are not comparable between laboratories

```
server/prisma/schema.prisma:617-633 ·
server/config/glosis_catalog.json (pSA procedures)
```

USDA places the silt/sand boundary at 0.05 mm; ISO 11277 and the FAO guidelines place it at 0.063 mm. Clay is 0.002 mm in both, so clay values are comparable — but a soil analysed under the two conventions will report meaningfully different silt and sand, with the difference falling where it matters most for texture class assignment near the loam boundaries.

The 25 GloSIS pSA procedures encode method and dispersion state, not the fraction scheme, so the ontology does not carry it either — which means the LIMS must, and it does not. For a platform whose stated purpose is harmonising seven national laboratories, this alone prevents the sand and silt columns from being pooled.

FIX

A required `fractionScheme` field (USDA / ISO-FAO / other, with explicit boundaries) on the particle-size analysis, carried on the result and emitted on export. Record it retrospectively per laboratory before pooling any existing data.

PSA-03 HIGH

No texture class is derived, so the one number clients actually read is missing

```
client/src/pages/DataResults.jsx:360 ·
server/controllers/dataResultsController.js:116 ·
client/src/components/report/ReportContent.jsx:206
```

Nothing in the codebase converts three fractions into a class name. "TEXTURE" appears as a filter that selects the three fraction codes, as an icon, and as a grouping label — never as a computation. The report's "Soil Texture" row is populated from `fieldMeta.soil_texture`: the surveyor's field estimate by hand-texturing, not the laboratory determination. So a certificate can print a field hand-texture of "sandy loam" beside laboratory fractions that

certificates can print a note naming texture of sandy loam. Record laboratory fractions that classify as clay loam, with nothing reconciling them.

GloSIS defines TextureLabClass as its own observation for the laboratory-determined class, and SandyTexture with its own value codelist. Both are unused.

FIX

Derive the class from the fractions on save, store it as its own result bound to TextureLabClass, and record which triangle was used (USDA and FAO/WRB differ in their class boundaries). Keep the field estimate as a separate, separately labelled row — the comparison between the two is diagnostically useful.

PSA-04 HIGH

The ± 0.5 % closure tolerance is too tight for the methods on offer

client/src/pages/TechWorkbench.jsx:21

TEXTURE_TOLERANCE = 0.5 applied uniformly. That is a defensible figure for a careful pipette determination where silt is obtained by difference and closure is near exact by construction. It is not defensible for the Bouyoucos hydrometer, where independent sand and clay readings routinely leave a 1–3 % residual, nor for laser diffraction, whose fractions are derived from an optical model with its own systematic offsets against sedimentation.

Two consequences, and the current code produces both at once: a tolerance this tight generates constant warnings on legitimate hydrometer work, which trains technicians to ignore the warning; and because the warning blocks nothing (PSA-01), genuinely bad sets pass alongside them. Standard laboratory practice is also being skipped — where closure falls within the method's acceptable window, fractions are *normalised* to sum to exactly 100 and the original sum retained, rather than being left inconsistent.

FIX

Tolerance per methodology, not a global constant. Normalise within tolerance and store both raw and normalised values; reject outside it.

PSA-05 MEDIUM

The fine-earth basis and coarse fragments are never stated or reconciled

server/prisma/schema.prisma (Sample, Result) · glosis_catalog.json (coarseFragments)

Particle size fractions are, by definition, percentages of the <2 mm fine earth. Coarse fragments are a separate determination on the whole sample. Nothing in the data model states which basis a result is on, and GRAVEL appears in the workflow icons as though it

were another fraction alongside sand, silt and clay – which, if a laboratory configures it that way, would put it into the same three-way sum and guarantee failure.

GloSIS carries coarseFragments with three procedures (field, field-from-class, laboratory sieved) and keeps it deliberately separate from pSA. Without the coarse-fragment fraction recorded separately, no volumetric conversion is possible – which is what blocks carbon stocks (see SCI-03).

FIX

An explicit basis attribute on every mass-fraction result (fine earth <2 mm / whole soil), and coarse fragments as a distinct property excluded from the texture sum.

04 The result record

Everything above rests on this table. Nine columns, of which four carry information: what a soil result needs, and what is actually stored.

A SOIL RESULT NEEDS	STORED?	CONSEQUENCE
Numeric value	String	Lexicographic sorting; no range queries without casting
Unit, controlled	Free text	Drives ADV-01; no QUDT mapping for GloSIS
Method actually used	No	Export and certificate cite the analysis default (GLO-03)
Analyst	No	On the work item only; lost at export and on the certificate
Date of analysis	updatedAt only	Cannot distinguish bench date from data-entry date
Instrument	No	Recorded on the work item, not the datum; no instrument recall
Replicate number	No	Duplicates cannot be stored at all (TRC-01)

LOD / LOQ, censoring	No	"<0.5" fails parsing and is stored as invalid text
Measurement uncertainty	No	No expanded uncertainty can be reported
Mass basis	No	Drives BAS-01
Depth / layer	No	No GloSIS feature of interest (SCI-02)
QC batch link	No	A failed batch cannot invalidate its results (QC-02)
Validation flags	Yes	Min/max only, single parameter

TRC-02 HIGH

A decimal comma is silently truncated rather than rejected

```
server/controllers/validationController.js:22 ·
client/src/components/report/ReportContent.jsx:65
```

Values are stored as strings and parsed with `parseFloat` wherever a number is needed. `parseFloat("12,5")` returns 12 — not NaN, not an error. It stops at the comma and returns what it has.

This platform serves laboratories in Guatemala, Honduras, Mozambique and Tunisia, and ships Spanish, Latin-American Spanish, Portuguese and French locales. The decimal comma is the norm in every one of them. A technician typing 12,5 on a numeric keypad produces a stored value of 12.5 as text that every downstream consumer reads as 12 — a 4 % error that passes validation, passes review, and prints on the certificate. For pH, entering 6,8 gives 6 rather than 6.8, moving the soil a full interpretation band.

FIX

Parse locale-aware at the boundary, store as a numeric column, and reject anything that does not parse cleanly and completely. Then sweep the existing `Result` table for values whose string form contains a comma.

TRC-03 HIGH

Values below the limit of quantification cannot be recorded

```
server/controllers/validationController.js:22-26 ·
resultsController.js:72
```

`validateResult` returns `{ valid: false, flags: ['INVALID_FORMAT'] }` for anything `parseFloat` cannot read — and then the caller upserts the row anyway, with `isValid: false`. So "`<0.5`" is stored as an invalid text value rather than as a censored measurement.

Left-censored results are unavoidable in soil analysis: micronutrients below the LOQ, total N in a sandy subsoil, available P in a strongly fixing andisol. They must be recorded with their limit so that downstream users can handle them properly, rather than being pushed into a free-text field where a consumer parsing with `Number()` gets NaN, or worse, where a technician substitutes a bare zero.

FIX

Separate value (numeric), censoring (none / below-LOQ / above-range) and `lod/loq` on the methodology.

TRC-04 HIGH

Every value on the certificate is printed to exactly two decimal places

`client/src/components/report/ReportContent.jsx:264`

`Number(item.value).toFixed(2)`, applied to every result regardless of parameter. This is wrong in both directions at once. A pH electrode is good to about 0.1 units and is conventionally reported so; printing `6.50` asserts a precision the measurement does not have. Texture fractions are reported to 0.1 % at best, and `34.00 %` is a claim no sedimentation method supports. In the other direction, a trace element near its detection limit — `0.003 mg kg-1` — is rendered `0.00` and the information is destroyed on the page.

Significant figures are not cosmetic in an analytical report. They are how a laboratory communicates the precision of its own method.

FIX

A `decimalPlaces` or significant-figure rule per analysis, set with the method, applied at render.

TRC-05 MEDIUM

Results live in two places that can disagree

`server/prisma/schema.prisma:150 (WorkItem.result) · Result table`

`WorkItem.result` holds a free-text copy of the value, and `Result` holds another. The workbench writes both in one transaction, but the deletion paths do not stay in step — several routines call `prisma.result.deleteMany` for a sample while the work-item copies survive, so a "cleared" sample can still show values in the work queue. Which of the two the

user sees depends on which screen they open.

FIX

One store. `WorkItem` should reference the result, not duplicate it.

05 Scientific completeness

What the model cannot express, in rough order of how much it constrains the programme's stated scientific goals.

SCI-02 HIGH

Sampling depth is not a field, and depth codes collide during Kobo sync

server/prisma/schema.prisma:82-126 ·
server/controllers/koboController.js:309, 468-472 ·
services/koboService.js:267-275

There is no depth column on `Sample`. Depth arrives from KoboToolbox as an opaque code — D1, D2 — inside the `fieldMetadata` JSON blob, alongside optional `depthTop` and `depthBottom` that the UI displays but nothing validates, types or requires.

A soil result without a depth interval is not a soil result. It cannot be compared to another laboratory's topsoil, cannot be mapped, cannot enter a stocks calculation, and cannot be expressed in GloSIS at all, because the feature of interest is a `GL_Layer` defined by its upper and lower depth.

There is a concrete operational hazard on top of the modelling one. The Kobo matcher normalises identifiers by *stripping* depth suffixes:

```
.replace(/[-_](?:D1|D2|T|S)$/i, '') // strip depth suffixes
```

So GTM-014-T (topsoil) and GTM-014-S (subsoil) normalise to the same key. Two physically different samples from one site compete for the same match, and field data can attach to the wrong depth.

FIX

First-class `depthTop` / `depthBottom` (cm, numeric, required for laboratory samples) plus an optional horizon designation, and include depth in the Kobo match key rather than

removing it.

SCI-01HIGH

Every multi-analyte family collapses the same way texture does

server/config/glosis_catalog.json · schema.prisma:617

Texture is the visible case because someone built a UI workaround for it. The same one-analysis-one-value constraint applies without any workaround to:

GLOSIS GROUP	PROCS	ANALYTES IN ONE DETERMINATION
exchangeableBases	11	Ca ²⁺ , Mg ²⁺ , K ⁺ , Na ⁺ from a single extraction
extractableElements	25	P, K, S and micronutrients from one Mehlich-3 or Olsen extract
totalElements	19	Full elemental suite from one digest
pSA	25	Sand, silt, clay from one dispersion

In each case one physical procedure on one aliquot yields several analytes that share a preparation, an instrument run, a calibration and a QC batch. Modelling each as an independent analysis multiplies the work items, breaks the shared QC linkage, and — because they all bind to the same GloSIS group — makes the analytes indistinguishable on export.

FIX

Introduce a composite analysis: one procedure, one work item, one equipment record, one QC membership, N result rows each bound to its own property. Texture becomes the first instance rather than a special case.

SCI-04HIGH

No derived quantities and no consistency checks between related parameters

server/controllers/validationController.js (whole file)

Validation is a per-parameter min/max and nothing else. Soil chemistry is full of internal constraints that a laboratory uses precisely because they catch errors no single-parameter range can

- Sum of exchangeable bases must not exceed CEC — a base saturation above 100 % is a calibration or dilution error, and it is the standard first check on a cation run.
- Base saturation, effective CEC and ESP are *derived* from bases and CEC. GloSIS carries baseSaturation and effectiveCec as properties; the system has no derivation engine, so they must be typed in by hand and can contradict their own inputs.
- Organic matter and organic carbon are related by a conversion factor (the 1.724 Van Bemmelen convention, or a locally calibrated one). Both are in the catalogue; nothing links them or records which factor was used.
- pH in KCl below pH in water by roughly 0.5–1.0 units is expected; the reverse indicates a net positive charge and is worth flagging, not silently accepting.
- Total N and organic C should give a plausible C:N ratio — a ratio of 3 or of 60 is almost always an error in one of the two.

FIX

A cross-parameter rule set evaluated server-side on submission, with rules declared per analysis group. This is the same machinery that texture closure needs, so build it once.

SCI-03 MEDIUM

Carbon stocks cannot be computed, which is the headline number the programme needs

`glosis_catalog.json` (bulkDensityFineEarth, bulkDensityWholeSoil, coarseFragments) · `schema.prisma`

A soil organic carbon stock requires organic carbon concentration, bulk density, the coarse fragment fraction and the depth interval, multiplied together. GloSIS carries all four — carbonOrganic (24 procedures), bulkDensityFineEarth (10), bulkDensityWholeSoil (8), coarseFragments (3). The LIMS can store the first as a number and has no structured home for the other three. For a soil-fertility programme reporting to a national or FAO audience, SOC stock in t C ha⁻¹ is usually the single most-requested output, and it is currently uncomputable from this database.

FIX

Falls out of SCI-02 (depth) and PSA-05 (coarse fragments) plus a bulk density analysis; then a derived quantity in the same engine as SCI-04.

SCI-05 MEDIUM**No sample preparation lineage, and no concept of an aliquot**

schema.prisma:95–99 (dryingStatus, preparationStatus) ·
workbenchController.js:519–530

Preparation is modelled as two boolean-ish gates on the sample — dryingStatus and preparationStatus — set when a DRYING or PREPARATION work item completes. That records that preparation happened, not what it produced.

In reality one field sample becomes several physically distinct materials: air-dried and sieved to 2 mm for most determinations; a further ball-milled subsample for total C and N; an undisturbed core for bulk density; sometimes a field-moist portion for mineral N. These have different masses, different storage and different validity windows, and a result belongs to one of them, not to the parent sample. Without an aliquot entity there is no way to record that the total carbon came from the milled fraction while the pH came from the 2 mm fraction — nor to trace a contamination problem to the mill.

FIX

An Aliquot entity between sample and result, with its preparation treatment. Larger than one sprint; worth scheduling.

SCI-06 MEDIUM**Preparation steps and measurements share one namespace**

schema.prisma:135 · workbenchController.js:519

WorkItem.analysis holds "DRYING, PREPARATION, PH_H2O, etc." — an operational gate and an analytical determination in the same field, distinguished by hard-coded string comparison in the controller. Drying is not an analysis: it has no property, no unit, no result, no method in the GloSIS sense, and it should not be able to receive a numeric value or appear in an analysis catalogue that is being bound to an ontology of measurements.

FIX

Separate the task type from the analysis code. The operational gates already have their own OperationalGate model — use it.

06 Quality control and spectroscopy

QC-01 HIGH**QC results are an unstructured JSON note, not evaluable data**

`server/prisma/schema.prisma:178-196`

The Batch model holds `qcResults String? // JSON string: { blanks: [], controls: [], duplicates: [] }` and `disposition String?`, both free-form. There is no certified reference material identity, no certified value, no acceptance window, no recovery percentage, no duplicate relative percent difference, no control limits, no warning/action rules, and therefore no control chart and no trend.

A QC system that cannot compute whether a control passed is a logbook, not a quality system. The structure implied by the comment is right; it just needs to be columns rather than a string.

FIX

Model CRM, blank, duplicate and spike as typed records with expected value, tolerance and computed outcome. Shewhart limits from the laboratory's own history follow naturally once the values are queryable.

QC-02 HIGH**A failed QC batch cannot invalidate the results it covers**

`schema.prisma:596-608 (no batchId on Result) · Batch.workItemIds`

Batch links to `WorkItem`; `Result` links to neither. When a batch is dispositioned `QC_FAIL`, there is no query that returns the results produced under it, and no mechanism to quarantine or flag them. The values remain in `Result`, indistinguishable from values produced under a passing batch, and they will be exported and reported.

Combined with `TRC-01` — a re-analysis overwrites rather than supersedes — a laboratory that detects a QC failure has no supported path to correct the record other than to overwrite it and lose the evidence that anything was wrong.

FIX

`batchId` on `Result`, and a disposition action that flags or withdraws every result in the batch.

SPC-01 MEDIUM**The spectral module archives spectra but predicts nothing**

```
server/controllers/spectralController.js ·  
services/spectralValidation.js · no prediction code found
```

Spectra are uploaded, QC-checked, reviewed and approved. There is no chemometric model, no calibration transfer, no prediction, and no path from a spectrum to a Result. The README's "spectral analysis" is, today, a spectral library.

This is a gap rather than an error, and the gap is currently protecting the data: GloSIS provides eight spectrally-predicted pSA procedures (pipette-disp-spec and siblings) specifically so that predicted values can be distinguished from measured ones, and since nothing is predicted, nothing is mislabelled. That protection ends the moment prediction is added — and with no provenance field on Result, a predicted clay content would be indistinguishable from a pipette determination.

FIX

Before any prediction lands: a provenance attribute on the result (measured / predicted / derived / imported), the model identity and version, and a prediction interval. Bind predicted values to the -spec procedures, which is exactly what they are for.

SPC-02 LOW

The MIR acceptance window excludes part of the fingerprint region

```
server/services/spectralValidation.js:15-18
```

MIR: { minWavenumber: 600, maxWavenumber: 4000 }. Mid-infrared soil spectrometers with KBr optics routinely collect to 400 cm⁻¹, and the 400–600 cm⁻¹ region carries Si–O and Al–O–Si lattice bands that are informative for clay mineralogy. A full-range scan will be flagged or trimmed at the boundary.

FIX

Widen to 400 cm⁻¹, or make the window a per-instrument property — which the equipment registry could already carry.

07 What is scientifically sound

Several judgements here are better than the surrounding implementation, and the remediation should build on them rather than around them.

- **The catalogue is real and correctly extracted.** 275 procedures with their notations, definitions, ISRIC references and full bibliographic citations, fetched from the authoritative repository and dated. That is the hard half of the raw material, and it is right.
- **The Methodology model is well designed.** Standard (ISO 10390:2021), procedure identity, notation, definition, reference, citation and a default flag, all per analysis. It is the correct shape for method management. Its only problem is that nothing at run time points at it.
- **The texture merge shows the right instinct.** Someone understood that sand, silt and clay are one determination and built the grouped entry row, the shared sum badge and the closure warning. The instinct is sound; it is implemented one layer too high.
- **Spectral QC is genuine.** Wavelength monotonicity, array length agreement, minimum point count, reflectance range with a deliberate allowance for overshoot above 1.0 — these are real checks written by someone who has seen bad spectra, not placeholders.
- **Operational gates are first-class.** Drying and preparation as tracked states with their own model and their own completion semantics reflects how a laboratory actually sequences work, and is more than most systems of this size bother with.
- **The certificate has the right anatomy.** Method and standard columns, methodology footnotes assembled from the used methods, interpretation bands with reference ranges, and a signature block. The structure is right; the content flowing into it is what needs correcting.

08 Remediation plan

Sequenced by consequence, not by difficulty. Phase A stops the system asserting things that are not true and can ship this week; everything after it is real modelling work.

PHASE A Stop the false assertions

2–3 days

Three claims currently leave the laboratory that the system cannot support. None of the fixes requires a schema change.

BAS-01

Remove the oven-dry sentence from the certificate footer, or replace it with the basis actually used. One line, today.

ADV-01	Suspend automatic interpretation until it is unit-aware. Print values and reference ranges without a verdict rather than printing a wrong verdict. Then re-enable per parameter as each band set is bound to a unit.
GLO-01	Delete both URI fallbacks in analysisController and sisController; export null for unmapped analyses. Audit the Analysis table and clear fabricated URIs.
TRC-02	Reject values that do not parse cleanly, and sweep existing results for comma decimals. Quantify the exposure before deciding how to correct them.
PSA-01	Make the browser's closure check actually block completion — an interim measure while the server-side rule is built, so the message stops being false.

PHASE B

Make the result record defensible

1–2 weeks

Everything else depends on a result knowing what it is. This is the highest-value structural work in the report.

TRC-01	Append-only results: methodologyId, replicateNo, isCurrent, supersededBy. Relax the unique constraint. Migrate existing rows as replicate 1, current.
TRC-01	Add enteredBy, analysedAt, equipmentId, batchId — all already known at the point of write in workbenchController, currently discarded.
TRC-02/03	Numeric value column with censoring, plus lod/loq and uncertainty on the methodology.
BAS-01	A basis attribute (air-dry / oven-dry / field-moist) and a moisture analysis whose value drives the correction. Then restore the footer sentence, truthfully.
TRC-04	Significant-figure rule per analysis, applied at render instead of the global toFixed(2).
QC-02	Link results to their QC batch and let a failed disposition flag them.

PHASE C

A real GloSIS binding

2 weeks

Separate property from procedure, and emit an observation a harvester could actually consume.

GLO-02	Extract the property codelists from glosis_layer_horizon.ttl into the catalogue
--------	---

alongside the procedures. Add `glosisProperty` to `Analysis`; keep `glosisProcedure` on `Methodology`. Re-map the existing analyses — a supervised exercise, not a script.

GL0-04	Bind units to QUDT (GM-PER-HectoGM, MOL-PER-KiloGM, ...) with a controlled picker replacing the free-text field. This is also what makes ADV-01 fixable properly.
GL0-03	Emit <code>usedProcedure</code> from the result's own methodology, and stop deriving method from the analysis default.
SCI-02	First-class depth interval on the sample, so the export has a <code>GL_Layer</code> to attach to. Fix the Kobo depth-suffix stripping at the same time.
GL0-05	Replace the ad-hoc <code>glosis</code> JSON block with a proper SOSA-shaped observation serialisation, and validate a sample export against the ontology before calling it conformant.
–	Update the admin screen's wording: these are procedures, not properties. The current text is teaching users the wrong model.

PHASE D

Composite analyses and derived quantities

2–3 weeks

One procedure, many analytes — the model that texture needed all along, applied to every family that has the same shape.

SCI-01	Composite analysis: one work item, one equipment record, one QC membership, N results each bound to its own property. Migrate texture to it first, then exchangeable bases and extractable elements.
PSA-01/04	Server-side closure with per-methodology tolerance; normalise within tolerance, store raw and normalised, reject outside. Store the sum against <code>SumTxtr_calcul</code> .
PSA-03	Derive the texture class, store it against <code>TextureLabClass</code> , record which triangle was applied, and show it beside the field estimate rather than instead of it.
PSA-02	Required fraction-scheme field (USDA / ISO-FAO). Backfill per laboratory before any cross-lab pooling.
SCI-04	Cross-parameter rule engine: bases $\leq$ CEC, C:N plausibility, pH-KCl below pH-H ₂ O, OM/OC factor. Same machinery as closure.
SCI-03	Bulk density and coarse fragments as proper analyses, then SOC stock as a derived quantity.

PHASE E

Quality system and method-aware interpretation

2-3 weeks

The parts that make the laboratory defensible to an assessor and useful to a farmer.

QC-01	Typed QC records — CRM with certified value and window, blank, duplicate with RPD, spike with recovery — and computed pass/fail. Control charts follow once the data is queryable.
ADV-01/02	Interpretation keyed on property, procedure and unit, with band sets per method and per country where the agronomy differs. Explicit refusal to interpret where